ECO AERA: Green Home Score & Retrofit Recommendation Engine

M2: Mid-submission | Megathon X: Singularity | Problem Statement 2 (set by Lloyds)

TEAM NUMBER: 274

1. Current approach and changes since M1

There are no changes to our approach since M1. ECO ARENA turns a home's property and energy details into a Green Home Score from 0 to 100, calculated by fixed rules: $S = w_1 s_1 + w_2 s_2 + ... + w_7 s_7$, where each sub-score s is out of 100 and the weights are efficiency 25%, consumption 20%, insulation 15%, heating 15%, glazing 10%, renewables 10% and smart controls 5%. The score falls into one of five bands (shown below). For homes rated Poor or Very poor, we suggest the retrofits that actually suit the property (insulation, heat pump, solar, double glazing, smart controls) in a sensible order, for example insulation before a heat pump, and estimate the electricity and gas saved, the carbon reduced and an indicative property-value range. A Retrofit Lab lets the user try upgrades and see the new numbers; the language model only explains results, never calculates them.

S = 0-29	S = 30-49	S = 50-64	S = 65-79	S = 80-100
Very poor	Poor	Moderate	Good	Highly efficient

2. Experiments run so far and their results

We began with the user-facing side. The frontend of the web application is built with **HTML, CSS and JavaScript** and follows the dashboard journey planned in M1. The flow below shows how a user moves through the pages that exist today.

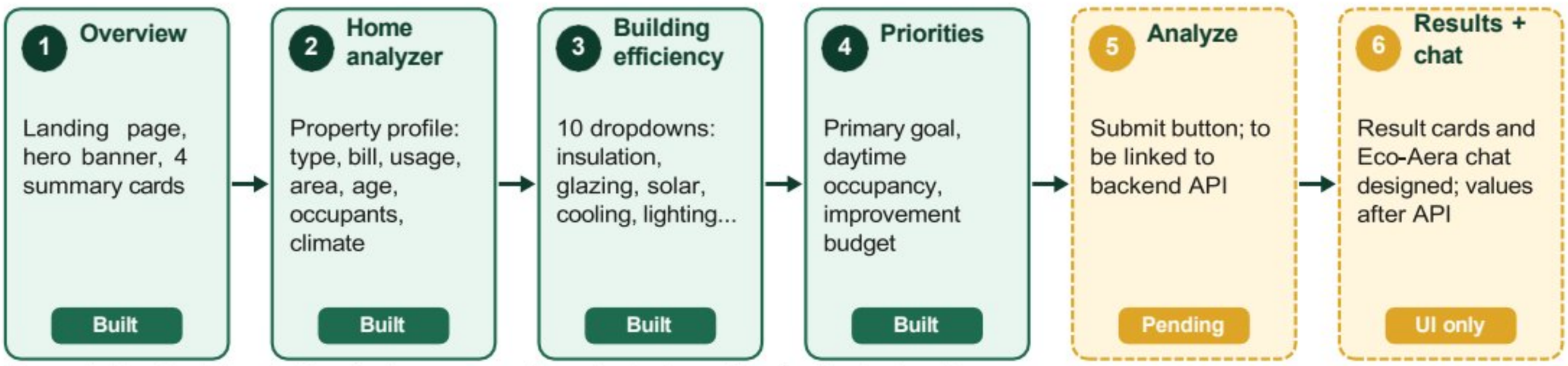
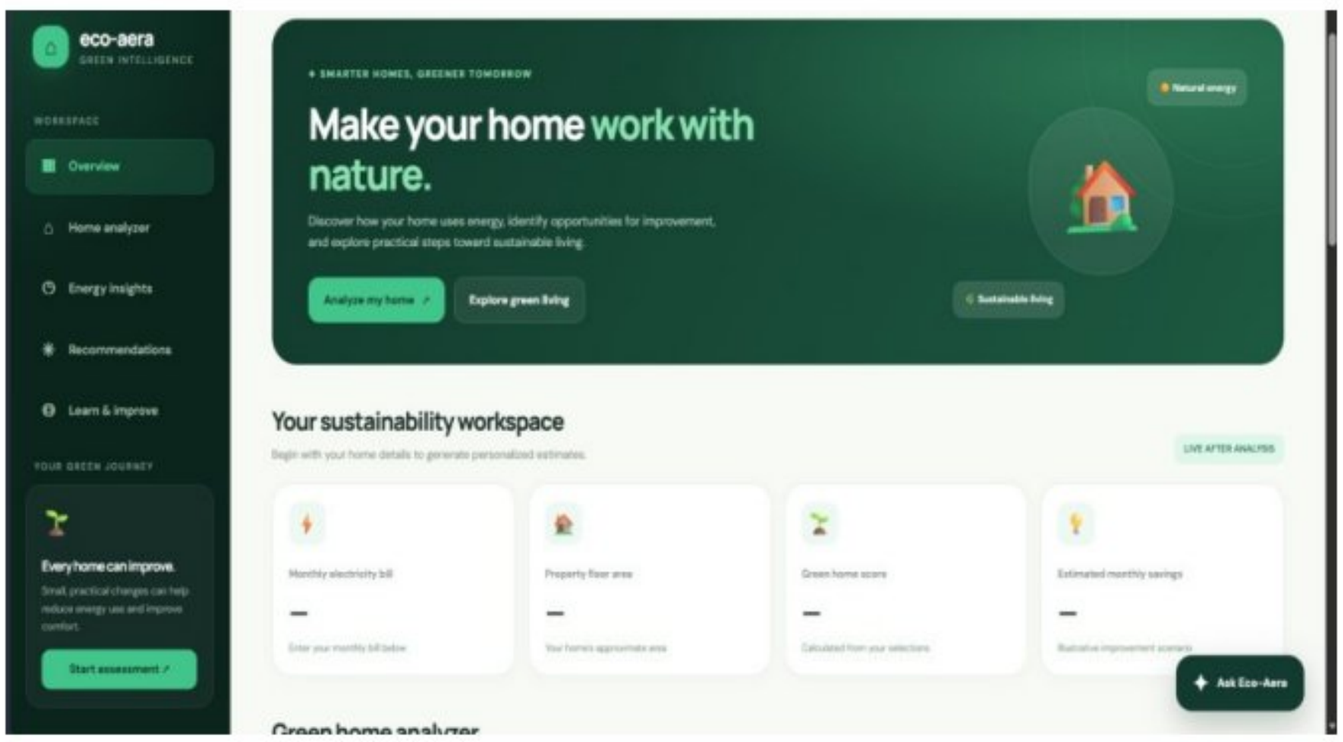
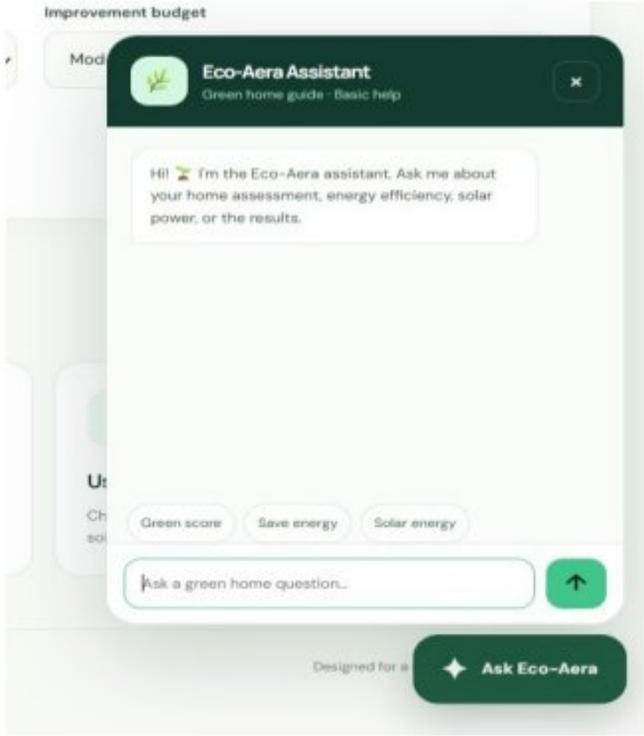


Figure 1: Frontend flow built so far (green = working, dashed = waiting for the backend).

Created Website :





What we built	Result
Layout and navigation	Sidebar with Overview, Home analyzer, Energy insights, Recommendations and Learn & improve; all sections open correctly.
Home analyzer form	Collects 7 property details, 10 building-efficiency choices and 3 improvement priorities, with a hint under each field.
Eco-Aera assistant	Chat window with quick prompts (Green score, Save energy, Solar energy) gives basic help; grounded LLM answers will follow.
Result cards	Bill, floor area, green score and savings cards are laid out; they show a dash until the backend is connected.

Scanned with OKEN Scanner

3. Progress on testing

The backend is being written in **Python**, and the data processing uses **NumPy** for unit harmonisation, calculating consumption per floor area and flagging missing values instead of filling them. We are testing each piece against hand-calculated cases before moving on, as planned in M1.

Part under test	How we check it
Green Home Score	Weights add up to 1, S stays between 0 and 100, band boundaries are correct, and the M1 worked example (sub-scores 40, 30, 50, 20, 60, 0, 0) gives S = 32.5, Poor.
Data processing	Unit conversion and kWh per floor area compared with manual calculation; missing or conflicting values are flagged, never silently invented.
Carbon calculation	Checked against hand calculation using emission factors stored with their units and source.

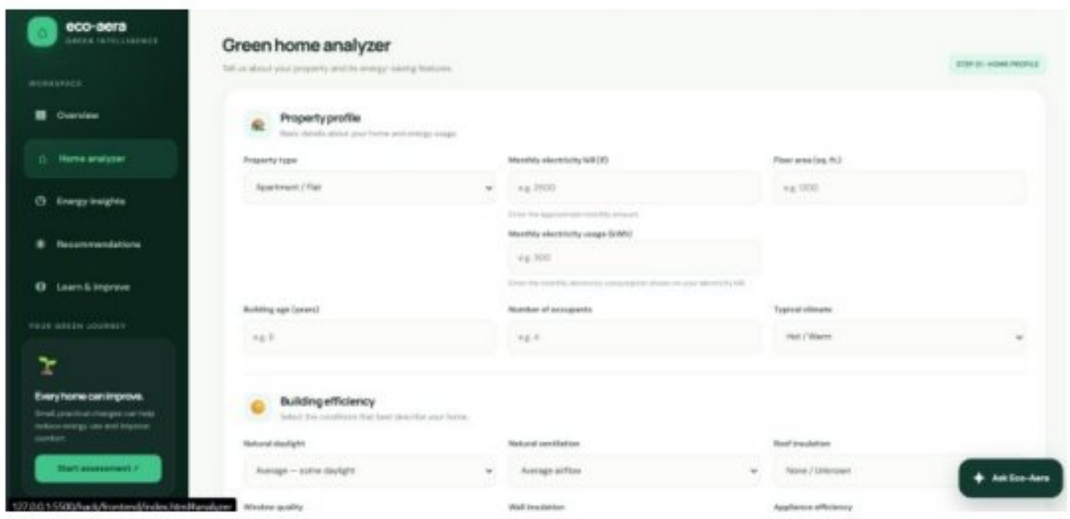
4. Plan for the remaining time

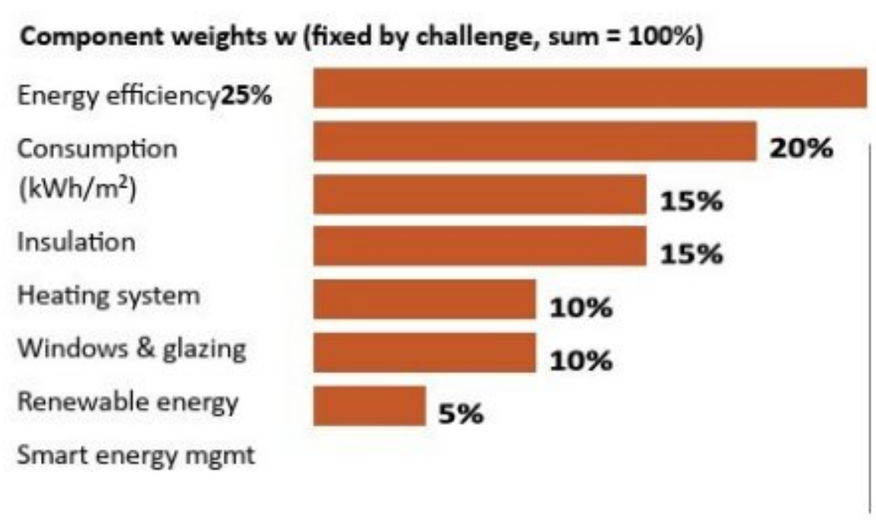
The next stage is to build the parts of the web application that need AI/ML, a database and an API, and join them to the frontend. Figure 2 shows where each block of our M1 pipeline stands today.



Figure 2: Project status by component.

Website & Calculated Output :





Score S	Band
80–100	Highly efficient
65–79	Good
50–64	Moderate
30–49	Poor
0–29	Very poor

Step	Work to be done
1. AI/ML models	Train gradient-boosted models for post-retrofit electricity and gas, property value and retrofit feasibility, each compared with a simple baseline.
2. Database	Store the synthetic property and energy data, emission factors with sources, and saved assessments.
3. API	Connect the Home analyzer to the backend so the score, recommendations and Retrofit Lab results fill the result cards.
4. LLM advisor	Make the Eco-Aera assistant answer only from computed results, and say 'unavailable' when data is not enough.
5. Final testing	Test on held-out homes, report validation and test results honestly, and push all code to main.

Scanned with OKEN Scanner